

{ English to Logic }

1- "Every file can be accessed by at least one manager"

$P(x)$: x is a manager, $f(x,y)$: x can access y

Logic: $\forall y \exists x (P(x) \wedge f(x,y))$

2- "There exist a manager who can access all files."

$P(x)$: x is a manager

$f(x,y)$ x can access file y

Logic: ~~$\forall x (P(x) \wedge \forall y f(x,y))$~~
 $\exists x (P(x) \wedge \forall y f(x,y))$

~~Logic:~~

S. R. K.

3 - Domain:
 u : all users in computer network
 f : all files on the server

Predicates:

$A(u)$: u is an administrator

$R(u, f)$: u can read file f

$W(u, f)$: u can write to file f

- Translate the following statements into English Logic:

"Every administrator can read every file"

Solution: $\forall u (A(u) \rightarrow \forall f R(u, f))$

"There exist a file that no admin can write to"

Solution: $\exists f \forall u (A(u) \rightarrow \neg W(u, f))$

"Every file can be read by at least one user"

Solution: $\forall f \exists u R(u, f)$

